Learning Fuzzy Descriptions from Crisp OWL Ontologies

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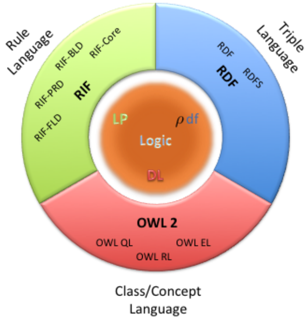
Overview

- Goal
- Background
- Learning Algorithms
- Evaluation
- Conclusions & Future Work

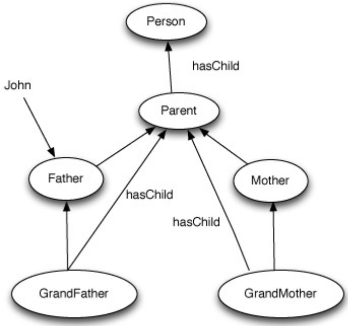
Goal

- Large amount of structured knowledge is nowadays available
- Wide variety of Semantic Web Languages has been used
  - RDFS: *Triple language*, *-Resource Description Framework*
    - The logical counterpart is ρdf
  - RIF: *Rule language*, *-Rule Interchange Format*,
    - Relates to *Logic Programming* (LP)
  - OWL 2: *Conceptual language*, *-Ontology Web Language*
    - Relate to Description Logics (DLs)
- Our focus: OWL 2



OWL 2

- OWL 2 is a decidable fragment of First-Order-Logic.
- The logical counterpart is the family of Description Logics
- OWL 2 in a nutshell: it allows to describe
  - classes
  - relations among classes
  - instances of classes



OWL 2 Profiles

OWL 2 EL
- Useful for large size of properties and/or classes
- Basic reasoning problems solved in polynomial time
- The EL acronym refers to the $\mathcal{EL}$ family of DLs

OWL 2 QL
- Useful for very large volumes of instance data
- Conjunctive query answering via query rewriting and SQL
- OWL 2 QL relates to the DL family *DL-Lite*

OWL 2 RL
- Useful for scalable reasoning without sacrificing too much expressive power
- OWL 2 RL maps to Datalog
- Computational complexity: same as for Datalog, polynomial in size of the data, EXPTIME w.r.t. size of knowledge base

Our Learning Problem

- Given
  - an OWL ontology
  - a target class T
  - a set of instances of class T

  induce class descriptions about T
- E.g., given a set of instances of Uncle and their age, learn

  "A person having a brother that is a father is an uncle"

- Improving readability:
  - induce *fuzzy class descriptions*

    "An old person having a brother that is a father is an old uncle"

  where being old is a function of a person's age.

Formal Background

From Crisp Sets to Fuzzy Sets.

- A classical set $A \subseteq X$ is characterised by the fact that either $x \in A$ or $x \notin A$
- Often sets are defined as
$$A = \{x \mid P(x)\}$$
  - $P(x)$ is a statement "x has property P"
  - $P(x)$ is either true or false for any $x \in X$
- Examples of universe X and subsets $A, B \in 2^X$ may be
$$\begin{array}{rcl} X & = & \{x \mid x \text{ is a person}\} \\ A & = & \{x \mid x \text{ is a father}\} \\ B & = & \{x \mid x \text{ is a father with age } a \geq 60\} \end{array}$$
- In the above case: $B \subseteq A \subseteq X$
- The membership function of a set $A \subseteq X$ is $\chi_A \colon X \to \{0, 1\}$, or simply
$$A \colon X \to \{0, 1\}$$
where $A(x) = 1$ iff $x \in A$
- Note that for sets $A, B \in 2^X$
$$A \subseteq B \text{ iff } \forall x \in X.\ A(x) \leq B(x)$$

- Complement of a set A, *i.e.* $\bar{A} = X \setminus A$: $\forall x \in X$:

$$\bar{A}(x) = 1 - A(x)$$

- Intersection and union: $\forall x \in X$

$$\begin{aligned} (A \cap B)(x) &= \min(A(x), B(x)) \\ (A \cup B)x) &= \max(A(x), B(x)) \end{aligned}$$

- Cartesian product of two sets $A, B \in 2^X$

$$A \times B = \{\langle a, b\rangle \mid a \in A, b \in B\}$$

- A relation $R \subseteq X \times X$
  - is reflexive if for all $x \in X$

$$R(x, x) = 1$$

  - is symmetric if for all $x, y \in X$

$$R(x, y) = R(y, x)$$

  - Inverse of R, $R^{-1} : X \times X \to \{0, 1\}$: $\forall x, y \in X$:

$$R^{-1}(y, x) = R(x, y)$$

- Fuzzy set: the dichotomy $x \in A$ or $x \notin A$ is no longer true
- The belonging of x to a fuzzy set A is graded
- The fuzzy membership function of a fuzzy set A is a function: $\chi_A : X \to [0, 1]$, or simply

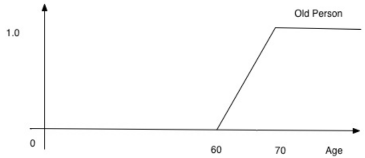
$$A : X \to [0, 1]$$

- Fuzzy power set over X, is denoted $\tilde{2}^X$, *i.e.* the set of all fuzzy sets over X
- Example: the fuzzy set

$$C \quad = \quad \{x \mid x \text{ is father whose age } a \text{ is old}\}$$

is defined via the membership function

$$\chi_C(x) = \begin{cases} 1 & \text{if } a \geq 70 \\ (x - 60)/10 & \text{if } a \in [60, 70) \\ 0 & \text{otherwise} \end{cases}$$



Standard Fuzzy Set Operations

- Standard fuzzy set operations:

$$\begin{aligned}(A \cap B)(x) &= \min(A(x), B(x)) \\ (A \cup B)(x) &= \max(A(x), B(x)) \\ \bar{A}(x) &= 1 - A(x) \ (\bar{A} \text{ is complement of } A)\end{aligned}$$

- *Inclusion degree* between A and B:

$$deg(A, B) = \frac{\sum_{x \in X} (A \cap B)(x)}{\sum_{x \in X} A(x)}$$

- *Cardinality* of a fuzzy set A:

$$|A| = \sum_{x \in X} A(x)$$

Norm-Based Fuzzy Set Operations

- Standard fuzzy set operations are not the only ones
- Most notable ones are triangular norms
  - t-norm $\otimes$ for set intersection
  - t-conorm $\oplus$ (also called s-norm) for set union
  - negation $\ominus$ for set complementation
  - implication $\Rightarrow$
    - set inclusion $A \subseteq B$ is defined as

$$\inf_{x \in X} A(x) \Rightarrow B(x)$$

    - $\Rightarrow$ is often defined from $\otimes$ as *r-implication*

$$a \Rightarrow b = \sup \{c \mid a \otimes c \leq b\} .$$

- These functions satisfy some properties that one expects to hold

Łukasiewicz, Gödel, Product Logic and Standard Fuzzy Logic

- One distinguishes three different sets of fuzzy set operations (called fuzzy logics)
  - Łukasiewicz, Gödel, and Product logic
  - Standard Fuzzy Logic (SFL) is a sublogic of Łukasiewicz
    - $\min(a,b) = a \otimes_l (a \Rightarrow_l b)$, $\max(a,b) = 1 - \min(1-a, 1-b)$

	Łukasiewicz Logic	Gödel Logic	Product Logic	SFL
$a \otimes b$	$\max(a+b-1, 0)$	$\min(a,b)$	$a \cdot b$	$\min(a,b)$
$a \oplus b$	$\min(a+b, 1)$	$\max(a,b)$	$a + b - a \cdot b$	$\max(a,b)$
$a \Rightarrow b$	$\min(1-a+b, 1)$	$\begin{cases} 1 & \text{if } a \leq b \\ b & \text{otherwise} \end{cases}$	$\min(1, b/a)$	$\max(1-a, b)$
$\ominus a$	$1-a$	$\begin{cases} 1 & \text{if } a = 0 \\ 0 & \text{otherwise} \end{cases}$	$\begin{cases} 1 & \text{if } a = 0 \\ 0 & \text{otherwise} \end{cases}$	$1-a$

- Mostert–Shields theorem: any continuous t-norm can be obtained as an ordinal sum of these three

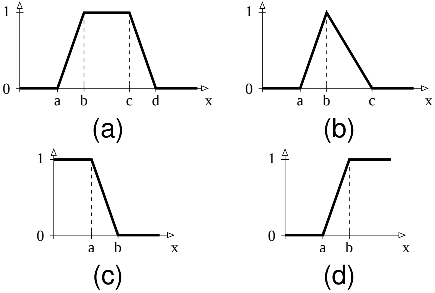
Some additional properties

Property	Łukasiewicz Logic	Gödel Logic	Product Logic	SFL
$x \otimes \ominus x = 0$	•			
$x \oplus \ominus x = 1$	•			
$x \otimes x = x$		•		•
$x \oplus x = x$		•		•
$\ominus \ominus x = x$	•			•
$x \Rightarrow y = \ominus x \oplus y$	•			•
$\ominus (x \Rightarrow y) = x \otimes \ominus y$	•			•
$\ominus (x \otimes y) = \ominus x \oplus \ominus y$	•	•	•	•
$\ominus (x \oplus y) = \ominus x \otimes \ominus y$	•	•	•	•

- Note: If all conditions in the upper part of a column have to be satisfied then we collapse to classical two-valued logic

On Fuzzy Membership functions

- Fuzzy membership functions may depend on the context and may be subjective
- Shape may be quite different
- Usually, it is sufficient to consider functions

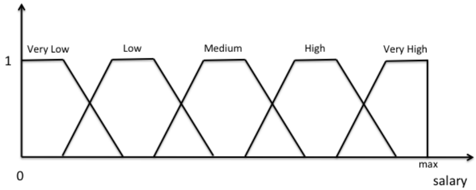


(a) Trapezoidal $trz(a, b, c, d)$; (b) Triangular $tri(a, b, c)$; (c) left-shoulder $ls(a, b)$; (d) right-shoulder $rs(a, b)$

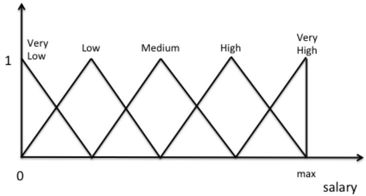
Fuzzy Sets Construction

- The usefulness of fuzzy sets depends critically on appropriate membership functions
- Methods for fuzzy membership functions construction is largely addressed in literature

- Easy and typically satisfactory method (numerical domain)
  - uniform partitioning into 5 fuzzy sets

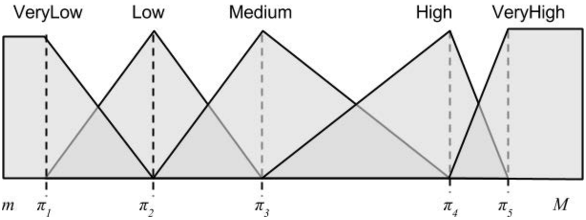


Fuzzy sets construction using trapezoidal functions



Fuzzy sets construction using triangular functions

- Another popular method is based on clustering
- Use Fuzzy C-Means to cluster data into 5 clusters
  - Fuzzy C-Means extends K-Means to accommodates graded membership
- From the clusters $c_1, \ldots, c_5$ take the centroids $\pi_1, \ldots, \pi_5$
- Build the fuzzy sets from the centroids



Fuzzy sets construction using clustering

Description Logics (DLs)

- Concept/Class: names are equivalent to unary predicates
  - In general, concepts equiv to formulae with one free variable
- Role or attribute: names are equivalent to binary predicates
  - In general, roles equiv to formulae with two free variables
- Taxonomy: Concept and role hierarchies can be expressed
- Individual: names are equivalent to constants
- Operators: restricted so that
  - Language is decidable and, if possible, of low complexity
  - No need for explicit use of variables
    - Restricted form of $\exists$ and $\forall$
  - Features such as counting can be succinctly expressed

- Basic ingredients: descriptions of classes, properties, and their instances, such as
  - a:C, meaning that individual a is an instance of concept/class C

$$\mathsf{I}a\text{:Person} \sqcap \forall \mathtt{hasChild}.\text{Femal}$$

  - (a, b):R, meaning that the pair of individuals $\langle a, b\rangle$ is an instance of the property/role R

$$(\mathsf{I}tom, \mathsf{I}mary)\text{:}\mathtt{hasChild}$$

  - $C \sqsubseteq D$, meaning that the class C is a subclass of class D

$$\text{Person} \sqsubseteq \forall \mathtt{hasChild}.\text{Person}$$

The (Fuzzy) DL Family

- A given DL is defined by set of concept and role forming operators
- E.g.: Fuzzy $\mathcal{ALC}(\mathbf{D})$ ($\mathcal{A}$ttributive $\mathcal{L}$anguage with $\mathcal{C}$omplement and concrete domain **D**), where $n \in [0, 1]$

Syntax				Semantics	Example
C, D	$\rightarrow$	$\top$	|	$\top(x)$	
		$\bot$	|	$\bot(x)$	
		A	|	$A(x)$	$Human$
		$C \sqcap D$	|	$C(x) \otimes D(x)$	$Human \sqcap Male$
		$C \sqcup D$	|	$C(x) \oplus D(x)$	$Nice \sqcup Rich$
		$\neg C$	|	$\ominus C(x)$	$\neg Meat$
		$\exists R.C$	|	$\exists y.R(x, y) \otimes C(y)$	$\exists has\_child.Blond$
		$\forall R.C$	|	$\forall y.R(x, y) \Rightarrow C(y)$	$\forall has\_child.Human$
		$\exists S.\mathbf{d}$	|	$\exists v.S(x, v) \otimes \mathbf{d}(v)$	$\exists has\_price. \geq_{120}$
		$\forall S.\mathbf{d}$		$\forall v.S(x, v) \Rightarrow \mathbf{d}(v)$	$\forall has\_age. =_{56}$
$\mathbf{d}$	$\rightarrow$	$ls(a, b)$	|		
		$rs(a, b)$	|		
		$tri(a, b, c)$	|		
		$trz(a, b, c, d)$	|		
		$\geq_v$	|		
		$\leq_v$	|		
		$=_v$	|		
		$C \sqsubseteq D$		$\forall x.C(x) \Rightarrow D(x)$	$Happy\_Father \sqsubseteq Man \sqcap \exists has\_child.Female$
		$a{:}C$		$C(a)$	$John{:}Happy\_Father$
		$\langle C \sqsubseteq D, n\rangle$		$\forall x.C(x) \Rightarrow D(x) \geq n$	$\langle CheapHotel \sqsubseteq GoodHotel, 0.7\rangle$
		$\langle a{:}C, n\rangle$		$C(a) \geq n$	$\langle John{:}Happy\_Father, 0.6\rangle$
		$\langle (a, b){:}R, n\rangle$		$R(a, b) \geq n$	$\langle (john, apple){:}likes, 0.9\rangle$

DL Knowledge Base

- A DL Knowledge Base is a pair $\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \rangle$, where
  - $\mathcal{T}$ is a TBox
    - containing general inclusion axioms of the form $C \sqsubseteq D$, $\langle C \sqsubseteq D, n \rangle$,
    - concept definitions of the form $A \doteq C$
    - primitive concept definitions of the form $A \sqsubseteq C$, $\langle A \sqsubseteq C, n \rangle$,
  - $\mathcal{A}$ is a ABox
    - containing assertions of the form $a{:}C$, $\langle a{:}C, n \rangle$
    - containing assertions of the form $(a, b){:}R$, $\langle (a, b){:}R, n \rangle$

Some Inference Problems

Consistency: Check if knowledge is meaningful

- Is $\mathcal{K}$ satisfiabile? $\mapsto$ Is there some model $\mathcal{I}$ of $\mathcal{K}$?

Subsumption: Structure knowledge, compute taxonomy

- $\mathcal{K} \models C \sqsubseteq D$? Is C a subclass of D?

Instantiation: Check if individual a is instance of class C

- $\mathcal{K} \models a{:}C$?

Retrieval: Retrieve set of individuals that instantiate C

- Compute the set $\{a \mid \mathcal{K} \models a{:}C\}$

Best Entailment Degree (BED): For $\tau \in \{C \sqsubseteq D, a{:}C, (a,b){:}R\}$

$$bed(\mathcal{K}, \tau) = \sup\{n \in [0,1] \mid \mathcal{K} \models \langle \tau, n \rangle\} .$$

Cardinality: The *cardinality* of a concept C w.r.t. a KB $\mathcal{K}$ and a set of individuals I

$$|C|^{\mathrm{I}}_{\mathcal{K}} = \sum_{a \in \mathrm{I}} bed(\mathcal{K}, a{:}C) .$$

Learning Algorithm

Setting.

Background theory. A consistent KB $\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \rangle$

Target concept. A class name T

Training set. $\mathcal{E} = \mathcal{E}^+ \cup \mathcal{E}^-$, set of positive and negative examples $a{:}T$ not entailed by $\mathcal{K}$

Goal. Find $\mathcal{H} = \{C_1 \sqsubseteq T, ..., C_n \sqsubseteq T\}$ (a *hypothesis*) such that:

Consistency. $\mathcal{K} \cup \mathcal{H}$ is a consistent;

Non-Redundancy. $\forall \tau \in \mathcal{H}, \mathcal{K} \not\models \tau$

Soundness. $\forall e \in \mathcal{E}^-, \mathcal{K} \cup \mathcal{H} \not\models e$;

Completeness. $\forall a{:}T \in \mathcal{E}^+, \exists i \in \{1, ..., n\}$ s.t. $\mathcal{K} \models a{:}C_i$

- Completeness is not always guaranteed to exist

Hypothesis Language. Fuzzy $\mathcal{EL}(\mathbf{D})$ inclusions $C \sqsubseteq T$

$$
\begin{array}{lcl}
C & \longrightarrow & \top \mid A \mid \exists R.C \mid \exists S.\mathbf{d} \mid C_1 \sqcap C_2 \\
\mathbf{d} & \rightarrow & ls(a,b) \mid rs(a,b) \mid tri(a,b,c) \mid trz(a,b,c,d) \,.
\end{array}
$$

CWA vs. OWA

How is $\mathcal{E}^-$ created?
Two different assumptions:

- *Open World Assumption (OWA)*:

$$\mathcal{E}^- = \{e \in \mathtt{Ind} \mid \mathcal{K} \models e : \neg T\} \; ;$$

- *Closed World Assumption (CWA)*:

$$\mathcal{E}^- = \{e \in \mathtt{Ind} \mid \mathcal{K} \not\models e : T\} \; ;$$

where $\mathtt{Ind}$ is the set of individuals appearing in $\mathcal{K}$.

Notice

Assuming either OWA or CWA, $\mathcal{E}^+ \cap \mathcal{E}^- = \emptyset$.

FOIL

Roughly, our learning algorithm proceeds as follows:

1. start from concept $\top$;
2. apply a refinement operator to find more specific concept description candidates;
3. exploit a scoring function to choose the best candidate;
4. re-apply the refinement operator until a good candidate is found;
5. iterate the whole procedure until a satisfactory coverage of the positive examples is achieved.

In FOIL

- each time a description is learned, the covered positive examples are removed from the training set
- the score of a candidate does not take into account previously learned descriptions
- the score of a candidate is based on a *GAIN* function

Refinement step (roughly):

- Add_A: add an atomic concept A
- $Add_{\exists R.\top}$: add a complex concept $\exists R.\top$ by existential role restriction
- $Add_{\exists T.\mathbf{d}}$: add a fuzzy concept $\exists T.\mathbf{d}$ by existential role restriction
- $Subst_A$: replace an atomic concept A with another atomic concept A' if A' subclass of A

$$
\text{REFINE}(C) = \begin{cases}
\mathcal{A} \cup \{\exists R.\top \mid R \in \mathcal{R}\} \cup \{\exists S.d \mid S \in \mathcal{S}, d \in \mathcal{D}\} & \text{if} \quad C = \top \\
\{A' \mid A' \in \mathcal{A}, \mathcal{K} \models A' \sqsubseteq A\} \cup \{A \sqcap A'' \mid A'' \in \rho(\top)\} & \text{if} \quad C = A \\
\{\exists R.D' \mid D' \in \rho(D)\} \cup \{(\exists R.D) \sqcap D'' \mid D'' \in \rho(\top)\} & \text{if} \quad C = \exists R.D, R \in \mathcal{R} \\
\{(\exists S.d) \sqcap D \mid D \in \rho(\top)\} & \text{if} \quad C = \exists S.d, S \in \mathcal{S}, d \in \mathcal{D} \\
\{C_1 \sqcap \ldots \sqcap C_i' \sqcap \ldots \sqcap C_n \mid i = 1, \ldots, n, C_i' \in \rho(C_i)\} & \text{if} \quad C = C_1 \sqcap \ldots \sqcap C_n
\end{cases}
$$

The FOIL Algorithm

function LEARN-SETS-OF-AXIOMS($\mathcal{K}$, T, $\mathcal{E}^+$, $\mathcal{E}^-$)
begin
$\mathcal{H} := \emptyset$; **D** = INITIALISEFUZZYCONCRETEDOMAIN($\mathcal{K}$);
while $\mathcal{E}^+ \neq \emptyset$ **do**
    $C \sqsubseteq T :=$ LEARN-ONE-AXIOM($\mathcal{K}$, T, $\mathcal{E}^+$, $\mathcal{E}^-$);
    $\mathcal{H} := \mathcal{H} \cup \{C \sqsubseteq T\}$;
    $\mathcal{E}^+_C := \{e \in \mathcal{E}^+ \mid \mathcal{K} \cup \{C \sqsubseteq T\} \models_+ e\}$;
    $\mathcal{E}^+ := \mathcal{E}^+ \setminus \mathcal{E}^+_C$;
endwhile
return $\mathcal{H}$
end

```
function Learn-One-Axiom(𝒦, T, ℰ⁺, ℰ⁻)
begin
C := ⊤;
φ := C ⊑ T;
ℰ⁻_φ := ℰ⁻;
while cf(φ) < θ or ℰ⁻_φ ≠ ∅ do
        C_best := C;
        maxgain := 0;
        𝒞 := REFINE(C)
        foreach C′ ∈ 𝒞 do
                φ′ := C′ ⊑ T;
                gain := Gain(φ′, φ);
                if gain ≥ maxgain then
                        maxgain := gain;
                        C_best := C′;
                endif
        endforeach
        φ := C_best ⊑ T;
        ℰ⁻_φ := {e ∈ ℰ⁻ | 𝒦 ∪ {φ} ⊨₊ e};
endwhile
return φ
end
```

Gain.

$$\text{GAIN}(\phi', \phi) = p * (log_2(cf(\phi')) - log_2(cf(\phi))) \ ,$$

p is the number of positive examples covered by the axiom ϕ that are still covered by ϕ'.

Confidence Degree. Fuzzy set inclusion degree between C and target T:

$$cf(\phi) = cf(C \sqsubseteq T) = \frac{\sum_{a \in \mathrm{Ind}^{+}_{\phi}(\mathcal{A})} bed(\mathcal{K}, a{:}C)}{|\mathrm{Ind}_{\phi}(\mathcal{A})|}$$

- $\mathrm{Ind}^{+}_{\phi}(\mathcal{A})$ (resp., $\mathrm{Ind}_{\phi}(\mathcal{A})$) is the subset of $\mathrm{Ind}(\mathcal{A})$ containing those individuals a involved in $\mathcal{E}^{+}_{\phi}$ (resp., $\mathcal{E}^{+}_{\phi} \cup \mathcal{E}^{-}_{\phi}$) such that $bed(\mathcal{K}, a{:}C) > 0$.

Fuzzy FOIL Classifier

- Suppose FOIL has induced a set of concept inclusions

$$\begin{array}{c} C_1 \sqsubseteq T \ , \\ \vdots \\ C_n \sqsubseteq T \ . \end{array}$$

- Suppose the confidence degree of these axioms are $cf_1, \ldots, cf_n$
- The FOIL classifier for target T is the weighted sum fuzzy inclusion axiom

$$cf_1 \cdot C_1 + ... + cf_n \cdot C_n \sqsubseteq T$$

pFOIL

Unlike FOIL, pFOIL

- Uses a probabilistic measure to score concept expressions
- Instead of removing positive examples covered from the training set as FOIL does, once an axiom has been learnt, positive examples are left unchanged after each learned rule
- Concept expressions are no longer evaluated one at a time, but the whole set of learnt expressions is evaluated as an ensemble

Probabilistic Fuzzy Ensemble Evaluation

For one rule $\mathcal{H} = \{C \sqsubseteq T\}$

- Fuzzy cardinalities

$$\begin{aligned} T^+ &= |T|^{\mathsf{I}_\mathcal{E}}_\mathcal{K} \\ C^+ &= |C|^{\mathsf{I}_\mathcal{E}}_\mathcal{K} \\ T^+ \sqcap C^+ &= |T \sqcap C|^{\mathsf{I}_\mathcal{E}}_\mathcal{K} \ . \end{aligned}$$

- Probability of correctly classifying a positive example

$$P(C^+ \mid T^+) = \frac{C^+ \sqcap T^+}{T^+} \ , \tag{1}$$

- Probability that a sample classified as positive is actually positive.

$$P(T^+ \mid C^+) = \frac{C^+ \sqcap T^+}{C^+} \ , \tag{2}$$

- Eq. 1 is *recall* of C w.r.t. T, Eq. 2 is *precision* of C w.r.t. T
- F_β – score function

$$score_\beta(C) = (1 + \beta^2) \cdot \frac{P(T^+ \mid C^+) \cdot P(C^+ \mid T^+)}{\beta^2 \cdot P(T^+ \mid C^+) + P(C^+ \mid T^+)}$$

General case: $\mathcal{H} = \{C_1 \sqsubseteq T, \ldots, C_n \sqsubseteq T\}$

Fuzzy Cardinalities.

$$\begin{aligned}\mathcal{H}^+ &= |C_1 \sqcup \ldots \sqcup C_n|^{\mathcal{E}}_{\mathcal{K}} \ ,\\ \mathcal{H}^+ \sqcap T^+ &= |(C_1 \sqcup \ldots \sqcup C_n) \sqcap T|^{\mathcal{E}}_{\mathcal{K}} \ ,\end{aligned}$$

Fuzzy Precision, Fuzzy Recall and Fuzzy F_β $-$ score score.

$$P(T^+ \mid \mathcal{H}^+) = \frac{\mathcal{H}^+ \sqcap T^+}{\mathcal{H}^+} \tag{3}$$

$$P(\mathcal{H}^+ \mid T^+) = \frac{\mathcal{H}^+ \sqcap T^+}{T^+} \tag{4}$$

$$score_\beta(\mathcal{H}) = (1+\beta^2) \cdot \frac{P(T^+ \mid \mathcal{H}^+) \cdot P(\mathcal{H}^+ \mid T^+)}{\beta^2 \cdot P(T^+ \mid \mathcal{H}^+) + P(\mathcal{H}^+ \mid T^+)} \tag{5}$$

Confidence Degree. Ensemble score of concept expression C w.r.t. $\mathcal{H} = \{C_1 \sqsubseteq T, \ldots, C_n \sqsubseteq T\}$

$$score_\beta(C, \mathcal{H}) = score_\beta(\mathcal{H} \cup \{C \sqsubseteq T\}) \ . \tag{6}$$

Stop Criterion.

- FOIL stops when all positive samples are covered or when it becomes impossible to find a concept without negative coverage.
- pFOIL stops a soon as the ensemble score improvement is below a threshold

Top-k Backtracking. At each step, instead of a single best candidate, a list of k best candidates is maintained

The pFOIL Algorithm

function LEARN-SETS-OF-AXIOMS($\mathcal{K}$, T, $\mathcal{E}^+$, $\mathcal{E}^-$, θ, β_1, β_2)
    $\mathcal{H} \leftarrow \emptyset$, **D** = INITIALISEFUZZYCONCRETEDOMAIN($\mathcal{K}$);
;
    oldScore $\leftarrow 0$;
    newScore $\leftarrow$ *score*$_{\beta_2}(\{\top \sqsubseteq T\})$;
    while *newScore* $-$ *oldScore* $> \theta$ **do**
        $C \leftarrow$ LEARN-ONE-AXIOM($\mathcal{K}$, T, $\mathcal{H}$, $\mathcal{E}^+$, $\mathcal{E}^-$, β_1);
        if(($C = \top$) **or** ($C =$ `null`))
            return $\mathcal{H}$;
        $\mathcal{H} \leftarrow \mathcal{H} \cup \{C \sqsubseteq T\}$;
        oldScore $\leftarrow$ *newscore*
        newScore $\leftarrow$ *score*$_{\beta_2}(\mathcal{H})$;
    return $\mathcal{H}$;
end

function LEARN-ONE-AXIOM($\mathcal{K}$, T, $\mathcal{H}$, $\mathcal{E}^+$, $\mathcal{E}^-$, β_1)

  $C \leftarrow \top$;

  while $|C|_{\mathcal{K}}^{\mathcal{E}^-} \neq 0$ **do**

    $C_{best} \leftarrow C$;

    $maxscore \leftarrow score_{\beta_1}(C_{best}, \mathcal{H})$;

    $\mathcal{C} \leftarrow$ REFINE(C);

    for all $C' \in \mathcal{C}$ **do**

      $score \leftarrow score_{\beta_1}(C', \mathcal{H})$;

      if $score > maxscore$ **then**

        $maxscore \leftarrow score$;

        $C_{best} \leftarrow C'$;

    if($C_{best} = C$)

      return null;

    $C \leftarrow C_{best}$;

  return C;

end

Top-k backtracking:

function LEARN-ONE-AXIOM($\mathcal{K}$, T, $\mathcal{H}$, $\mathcal{E}^{+}$, $\mathcal{E}^{-}$, β_1)
$C \leftarrow \top$;
$stack \leftarrow \emptyset$;
while $|C|_{\mathcal{K}}^{\mathcal{E}^-} \neq 0$ **do**
$C_{best} \leftarrow C$;
$maxscore \leftarrow score_{\beta_1}(C_{best}, \mathcal{H})$;
$\mathcal{C} \leftarrow$ REFINE(C);
for all $C' \in \mathcal{C}$ **do**
$score \leftarrow score_{\beta_1}(C', \mathcal{H})$;
ADD(C', $score$, $stack$); ,
if $score > maxscore$ **then**
$maxscore \leftarrow score$;
$C_{best} \leftarrow C'$;
if $C_{best} = C$ **then**
if $stack \neq \emptyset$ **then**
$C_{best} \leftarrow$POP($stack$);
else return `null`;
$C \leftarrow C_{best}$;
return C;
end

Evaluation

Setup.

- Prototype implemented comparing FOIL vs. pFOIL in a classification problem
- Various OWL ontologies have been considered
- For each ontology a meaningfull target concept T has manually been selected
- Learning algorithm has been applied and effectiveness measured
  - Negative examples $\mathcal{E}^-$ is the set of the individuals occurring in the KB which cannot be proved to be instance of T (*Closed World Assumption*
  - 5-fold cross validation
  - Performance indices: *Fuzzy Precision* (denoted P), *Fuzzy Recall* (denoted R), fuzzy $F1$-score (denoted $F1$) and the *Mean Squared Error* (MSE), defined as

$$MSE = \sum_{a \in \mathrm{I}_{\mathcal{E}}} (\mathcal{H}(a) - \mathcal{E}(a))^2 \ ,$$

where $\mathcal{E}(a) = 1$ if a is a positive example (*i.e.*, $a \in \mathrm{I}_{\mathcal{E}+}$), $\mathcal{E}(a) = 0$ if a is a negative example (*i.e.*, $a \in \mathrm{I}_{\mathcal{E}-}$), and $\mathcal{H}(a)$ is the degree of being a instance of T, defined as

$$\mathcal{H}(a) = bed(\mathcal{K} \cup \mathcal{H}, a{:}T) \ .$$

ontology	**DL**	**class**	**obj. p.**	**data. p.**	**instan.**	**target**	max d./c.
Family-tree	$\mathcal{SROIF}(\mathcal{D})$	22	52	6	368	Uncle	2/5
Hotel	$\mathcal{ALCOF}(\mathcal{D})$	89	3	1	88	Good_Hotel	2/5
Moral	$\mathcal{ALC}$	46	0	0	202	Guilty	2/5
SemanticBible	$\mathcal{SHOIN}(\mathcal{D})$	51	29	9	723	Woman	2/5
UBA	$\mathcal{SHI}(\mathcal{D})$	44	26	8	1268	Good_Researcher	2/5
FuzzyWine	$\mathcal{SHI}(\mathcal{D})$	178	15	7	138	DryWine	2/5

Ontology	Example of **induced axiom**
Family-tree	$(\mathtt{Person} \sqcap (\exists \mathtt{ancestorOf}.\top)) \sqsubseteq \mathtt{Uncle}$
Hotel	$\mathtt{Bed\_and\_Breakfast} \sqcap (\exists \mathtt{hasPrice}.\mathtt{High}) \sqsubseteq \mathtt{Good\_Hotel}$
Moral	$\mathtt{blameworthy} \sqsubseteq \mathtt{Guilty}$
SemanticBible	$\exists \mathtt{spouseOf}.\mathtt{Man} \sqsubseteq \mathtt{Woman}$
UBA	$\exists \mathtt{hasNumberOfPublications}.\mathtt{VeryHigh} \sqsubseteq \mathtt{Good\_Researcher}$
FuzzyWine	$\mathtt{OffDryWine} \sqcap \mathtt{SweetWine} \sqsubseteq \mathtt{DryWine}$

Ontology	**P**	**R**	**F1**	**MSE**	*t*	
Family-tree	1.0	1.0	1.0	0.0	18.05	FOIL
	1.0	1.0	1.0	0.0	161.38	pFOIL
Hotel	0.2855	0.1211	0.1696	0.0626	1.00	
	0.4223	0.3211	0.2845	0.0536	3.27	
Moral	1.0	1.0	1.0	0.0	0.85	
	1.0	1.0	1.0	0.0	27.41	
SemanticBible	0.4300	0.1390	0.2038	0.0311	5.20	
	0.4408	0.7729	0.5524	0.0203	8.44	
UBA	1.0	0.9453	0.9683	0.0005	0.44	
	1.0	0.9600	0.9778	0.0004	1.46	
FuzzyWine	0.9267	0.9500	0.9310	0.0110	1.25	
	0.9267	0.9500	0.9310	0.0110	35.1	

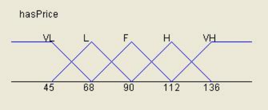
The Hotel Case Study

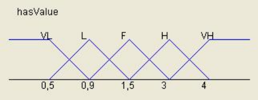
What are Good hotels, using TripAdvisor data?

- OWL 2 Ontology about meaningful city entities and their descriptions (the Hotel ontology)
- TripAdvisor data about hotels and user judgments of hotels in Pisa, Italy
- Examples rules learned for *e.g.*, Pisa, Italy

$$
\begin{array}{lll}
\mathit{Hostel} & \sqsubseteq & \mathit{Good\_Hotel} \\
\mathit{Hotel} \sqcap \exists \mathit{hasPrice}.\mathit{VeryHigh} & \sqsubseteq & \mathit{Good\_Hotel} \\
\mathit{Hotel} \sqcap \exists \mathit{hasPrice}.\mathit{High} & \sqsubseteq & \mathit{Good\_Hotel}, \quad \mathit{cf} = 0.569 \\
\mathit{Hotel} \sqcap \exists \mathit{hasDistance}.(\exists \mathit{isDistanceFor}.\mathit{Bus\_Station}) \sqcap \exists \mathit{hasValue}.\mathit{Low})) & & \\
\sqcap \exists \mathit{hasDistance}.(\exists \mathit{isDistanceFor}.\mathit{Town\_Hall}) \sqcap \exists \mathit{hasValue}.\mathit{Fair}) & & \\
\sqcap \exists \mathit{hasPrice}.\mathit{VeryLow} & \sqsubseteq & \mathit{Good\_Hotel}, \quad \mathit{cf} = 0.739 \\
\mathit{Hotel\_3\_Star} & & \\
\sqcap \exists \mathit{hasDistance}.(\exists \mathit{isDistanceFor}.\mathit{Train\_Station}) \sqcap \exists \mathit{hasValue}.\mathit{VeryLow})) & & \\
\sqcap \exists \mathit{hasPrice}.\mathit{Fair} & \sqsubseteq & \mathit{Good\_Hotel}, \quad \mathit{cf} = 0.289
\end{array}
$$

- Real valued price attribute *hasPrice* has been automatically fuzzyfied





Conclusions & Future Work

- We have presented two FOIL-based algorithms, FOIL and pFOIL, that, given
  - an OWL ontology
  - a target concept T

  induces (fuzzy) $\mathcal{EL}(D)$ concept descriptions about T
- pFOIL is based on
  - FOIL
  - a probabilistic fuzzy ensemble estimation
  - top-k backtracking
- pFOIL is better than FOIL in effectiveness so far, but is slower in running time.

- Future Work: working out fuzzy versions of other popular methods
    - Decision Trees
    - Hybrid Learning (genetic algorithms)
        - Already implemented. Results are discouraging so far.
    - Ensemble-based learners
        - AdaBoost (under preparation)
        - Random Forest
        - Random FOIL
- Fuzzy Sets Parameter Learning